HASNAE ELBALLAOUI

AI & Digital Transformation Engineering Student
+212 781-863989 | hasnaeelballaoui@gmail.com | Benguerir, Morocco
linkedin.com/in/hasnae-elballaoui-312209256 | github.com/Hasnae112 | Portfolio: hasnaeelballaoui.vercel.app

Seeking a final-year internship (PFE) in Artificial Intelligence starting February 2027

PROFILE

Fifth-year engineering student specializing in Artificial Intelligence and Digital Transformation, with hands-on experience in end-to-end AI, Machine Learning, and Data Engineering projects. Skilled in Python, SQL, data preprocessing, feature engineering, predictive modeling, model evaluation, and ETL pipeline development. Combines strong analytical thinking with practical problem-solving and a focus on building reliable, data-driven solutions. Seeking a PFE internship starting February 2027.

EDUCATION

École Nationale des Sciences Appliquées d'Al Hoceïma (ENSAH) *2022 – Present*

Engineering Degree – Artificial Intelligence and Digital Transformation (preceded by an integrated preparatory cycle)

Main courses: Cloud Computing, Machine Learning, Deep Learning, NLP, Cybersecurity, DevOps, Data Engineering, Software Engineering

PROFESSIONAL EXPERIENCE

PFA Intern – NLP / AI | 3D Smart Factory (Mohammedia, remote) *7/2026 – 9/2026*

Deep, personalized hybrid pipeline for candidate–job matching

- CV–job matching pipeline: Sentence-Transformers embeddings, FAISS index, Cross-Encoder re-ranking, skills-oriented score and personalization.
- Evaluation on 344 CVs and 28 offers: NDCG@5 of 0.900 vs 0.849 (BM25) and 0.790 (random); diagnosis and correction of an initially counter-productive Cross-Encoder.
- Streamlit application, job collection via APIs (Remotive, France Travail), partial Airflow/Docker orchestration.

Web Developer Intern | ALKHEBRA PRO *8/2025 – 9/2025*

- Design and development of the company's complete website (HTML, CSS, JavaScript), responsive interface.
- Full cycle handled: structure, interface, implementation and testing.

ACADEMIC PROJECTS

Software Engineering Chatbot – NLP Comparative Study (GitHub) *Python, PyTorch, FLAN-T5, LoRA, FAISS*

- Comparison of three architectures: Seq2Seq with attention, fine-tuned FLAN-T5 (LoRA) and RAG.
- RAG (FAISS, BM25, Cross-Encoder) selected: more effective with fewer hallucinations; evaluated with BLEU, ROUGE, MRR.

Job Intelligent – Job-offer Data Pipeline (GitHub) *Python, Airflow, MinIO, PostgreSQL, Docker*

- ETL pipeline unifying 4 job sources (Kaggle, France Travail, Rekrute, Adzuna), orchestrated with Airflow and containerized with Docker.
- MinIO data lake (Bronze, Silver, Archive, Gold) and PostgreSQL star-schema data warehouse, Power BI reporting.

CAN 2025 – Winner Prediction (GitHub) *Python, Scikit-Learn, Random Forest*

- Merged 5 sources into a base of 8,297 matches; 20 engineered features (FIFA ranking, form, fatigue, head-to-head).
- Tuned Random Forest (minimal overfitting), then 1,000 Monte Carlo simulations of the full tournament.

ATTENDIFY – Face-recognition Attendance (GitHub) *Python, OpenCV, LBPH, Tkinter, MySQL*

- Desktop app: face detection (Haar Cascade) and recognition (LBPH), student management, MySQL storage and CSV export.

GestionPFE – Final-year Project Defense Scheduling (GitHub) *Java, Spring MVC, Apache POI*

- Web app: Excel import and constraint-based schedule generation (max 4 defenses per professor per day, 1-hour break, unavailability), validation and Excel/Word exports.

Conversational IoT Assistant – Cloud Deployment (GitHub) *Python, Bot Framework, Azure, GitHub Actions*

- Chatbot controlling simulated devices, deployed to Azure Web App with a GitHub Actions CI/CD pipeline.

Smart Home IoT Security – MQTT Cybersecurity (GitHub) *MQTT, TLS, Node-RED, Python, Docker*

- IoT cybersecurity project for a smart home: demonstration of MQTT vulnerabilities and implementation of protections (TLS, authentication, ACL).
- Integration with Node-RED and Python, containerized with Docker.

SKILLS

Programming languages: Python, Java, SQL, PHP, HTML, CSS, JavaScript
AI & Machine Learning: Machine Learning, Deep Learning, NLP, LLMs & RAG, Computer Vision, Scikit-Learn, PyTorch, Pandas, NumPy, OpenCV, Sentence-Transformers, FAISS, Cross-Encoders, BM25
Tools & environment: Linux, Git/GitHub, Docker, Airflow, Hadoop, Spark, MinIO, Flask, Streamlit, Odoo, PostgreSQL, MySQL, Power BI, ETL pipelines, Agile (Scrum)
Soft skills: Curiosity, autonomy, analytical mind, teamwork, problem solving, adaptability, attention to detail

LANGUAGES

Arabic: Native | French: Fluent | English: Advanced